

MUDITHA PRIYASAD

Mechanical Design, Automation & R&D Engineer

IESL Associate Member (AM-32795)

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PROFESSIONAL SUMMARY

Mechanical Design, Automation & R&D Engineer with 2+ years' experience across 15+ delivered projects spanning machine design, prototyping, commissioning and troubleshooting of production equipment. Skilled in PLC-based control, pneumatics, DfMA and rapid prototyping, with a track record of delivering low-cost automation and tooling that measurably improve throughput, consistency and operator productivity.

CORE SKILLS

- **Mechanical Design:** mechanisms, jigs & fixtures, 3D CAD, DfMA, fabrication support
- **Industrial Automation:** PLC programming, sensors, actuators, pneumatics, commissioning
- **Manufacturing Improvement:** root-cause analysis, TPM, 5S, Kaizen, SOPs, KPI tracking
- **Prototyping & R&D:** 3D printing, CNC laser engraver, rapid prototyping, design validation
- **Professional Skills:** Cross-functional Collaboration, Continuous Improvement, Design Thinking
- **Tools:** SOLIDWORKS, Siemens STEP 7 Micro/WIN, Delta ISPSOFT, Mitsubishi GX Works, Xinje PLC Software, MS Excel, OrcaSlicer, Cura, LightBurn, Inkscape, Arduino IDE, Python, C++, Raspberry Pi
- **Also Familiar With:** AutoCAD, MATLAB, Power BI

PROFESSIONAL EXPERIENCE

MARQO Industries — Mechanical Design & Automation Engineer

Dec 2024 – Present

- Appointed as the company's first engineer, building the engineering and automation function from the ground up — workshop setup, tooling, inventory systems and department software.
- Designed and commissioned production machines, jigs, fixtures and pneumatic mechanisms to reduce manual effort and improve productivity.
- Set up, commissioned and troubleshoot spot welding, packing, pad printing and LED cup installation machines.
- **Semi-automatic fan base filling machine** – increased throughput from 1 to 6 bases/min with operator loading/refilling.
- **B22 eyelet puncher** – increased throughput from ~3–4 to 16 pcs/min while reducing operator effort.
- **1W LED bulb assembly machine (prototype)** – semi-automatic design with capacity up to 20 pcs/min.
- **Fan motor assembly fixture** – hybrid 3D-printed and laser-cut stainless steel design, saving approx. 5 seconds per motor across 50–80 motors/day while reducing operator strain.

MTone Designers — Founder

Jun 2021 – Present

- Delivered 20+ custom jobs across 3D printing, laser engraving, wood-burning gifts and customised engineering work.
- Built and tested 3 custom 3D printers and 4 laser machines for prototyping, production and personal/customer projects.
- **PCB drilling machine (NDM Electrical)** – automated design enabling 18 PCBs per setup using stacked PCB positioning.
- **Pneumatic wire-stripping tool** – developed for customised part production alongside 3D printing and laser engraving work.

MAS Active Shadowline — Engineering Intern

Apr 2024 – Nov 2024

- Supported TPM, 5S, Kaizen and KPI tracking; acted as interim Sustainability Executive to maintain reporting.
- Prepared operational checklists, machine standards and designed a sewing-machine repair workshop layout.

Effective Engineering & Automation (Pvt) Ltd — Automation Engineering Intern *Apr 2023 – Oct 2023*

- Assisted client projects: tensile testing, concrete mixer, oil filling, glove-making machine service and screen-printer troubleshooting.
- **Custom 3D printer** – designed, assembled and validated (mechanical, electronics, firmware).
- **Self-tightening tensile-test gripper** – replaced the screw-guided clamp for low-force rubber testing.

MAS Kreedaa Vaanavil — Autonomation Engineering Intern

Feb 2019 – Apr 2019

- Applied CAD, wiring, pneumatics, Arduino and PLC ladder logic to machine development.
- Contributed to bonding tape automation, fabric-picking research and a remote boiler monitoring case study.

PROJECTS

- Combat Robots | Line follower with liftable gripper | Align Tank | TPM Implement project | MTone ERP Software (Photos, drawings and videos available in portfolio.)

EDUCATION

BSc (Hons) Mechanical Engineering — University of Jaffna

Oct 2017 – Feb 2024

- Final-year research: grasshopper-inspired jumping mechanisms; designed and fabricated a two-actuator hopping robot.
- Led combat robot team; built an 8x8x8 LED cube and held leadership roles in practical projects and exhibitions.

CERTIFICATIONS

- CSWA — Mechanical Design
- CSWA-AM — Additive Manufacturing
- PLC Training — University of Moratuwa (2025)
- Maintenance of Industrial Plant & Machinery — NERDC (2026)
- Industrial Motor Operation & Control — NERDC (2025)
- Advanced Injection Moulding — IDB (2026)

ACHIEVEMENTS & ACTIVITIES

- Bronze Award — 3rd place for university stall with the best display of engineering projects, Techno Sri Lanka (2023)
- Participant — Air Tattoo Exhibition (2024); Techno Sri Lanka (2023); SLT Robot Battles (2019)
- Organiser — UOJ Battlegrounds (2020)

LANGUAGES

- English — professional proficiency
- Sinhala — native